



DESIGN AND ANALYSIS OF MACHINE LEARNING EXPERIMENTS

Presented By

J.Kowsy Sara
Assistant Professor
AI&ML

Rathinam Technical Campus
kowsysara.soc@rathinam.in

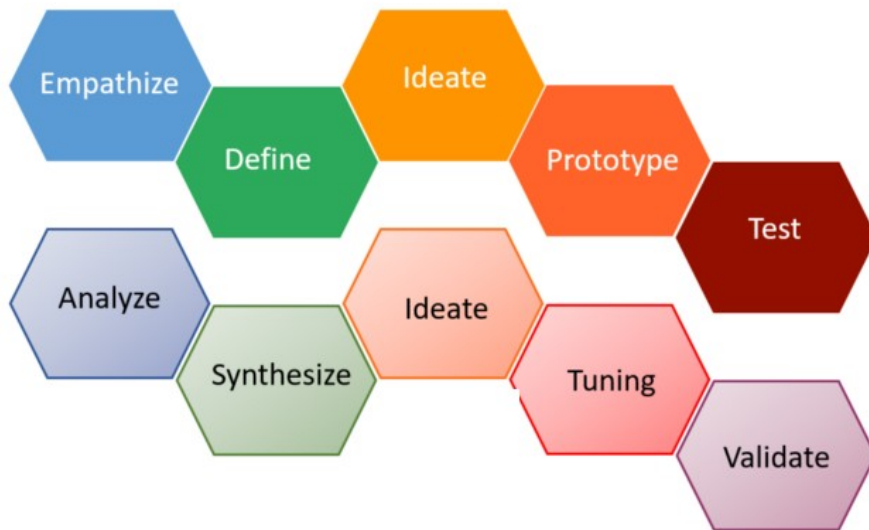
DESIGN AND ANALYSIS OF MACHINE LEARNING EXPERIMENTS

Design and Analysis of Machine Learning Experiments involves planning how to train, validate, and test models to ensure reliable and reproducible results.

Design Thinking
designs products this way...

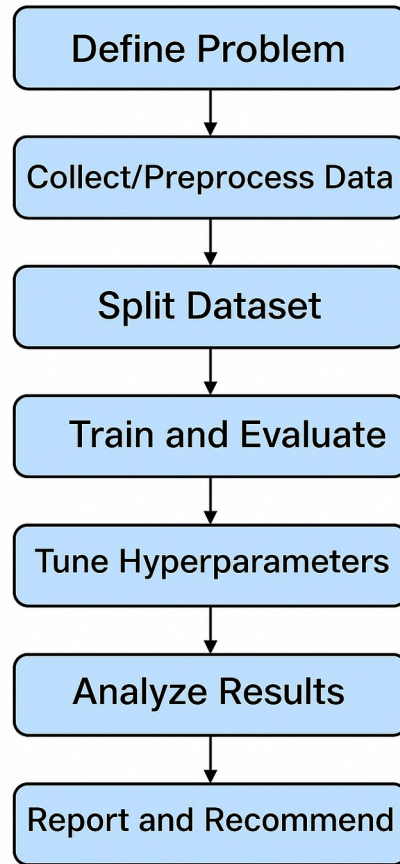
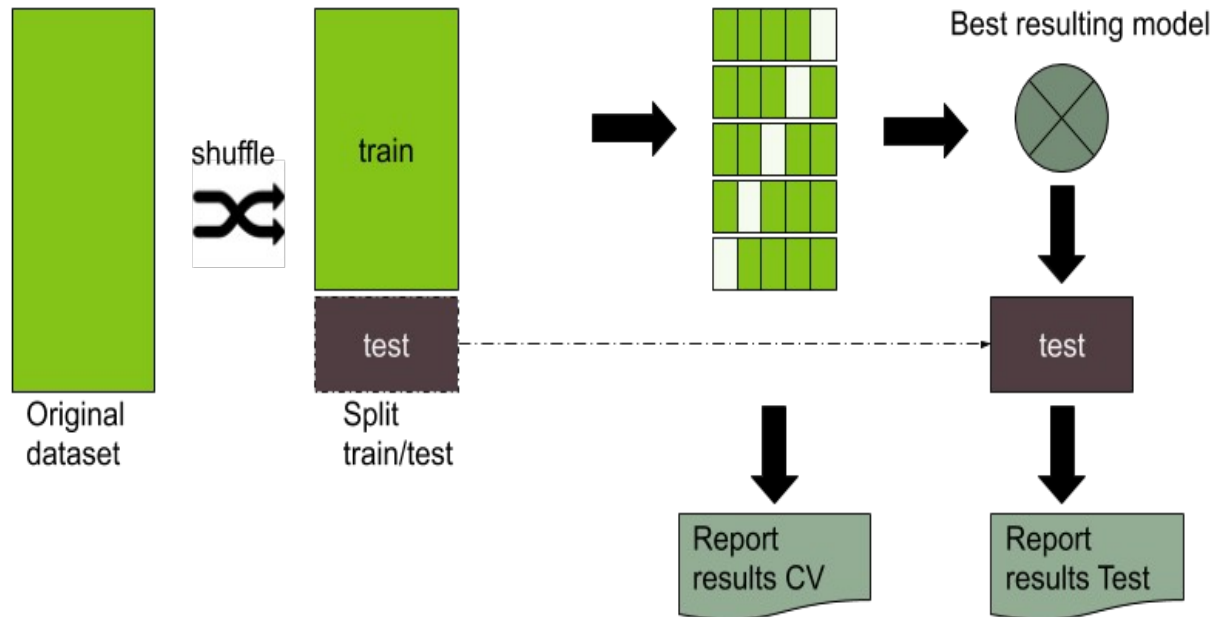
refines, tunes and predicts this way...

Machine Learning



DESIGN AND ANALYSIS OF MACHINE LEARNING EXPERIMENTS

EXPERIMENTAL DESIGN A



1. Guidelines for Machine Learning Experiments

Definition: Steps to properly plan, run, and analyze ML models to ensure reliable and fair evaluation.

Key Guidelines:

- Use **representative data** from the real-world problem.
- Separate data into **training**, **validation**, and **test** sets.
- **Avoid overfitting** by not testing on training data.
- Use **resampling techniques** like cross-validation for better reliability.
- **Repeat experiments** and **report average performance**.

Real-Time Example:

In **credit card fraud detection**, you should train on past transactions, validate on unseen ones, and test on recent real data. Report average accuracy over different samples to confirm stability.

Cross Validation (CV) and resampling

•**Sampling**: Collecting the initial data from a population (e.g., survey, sensors, transaction logs).

•**Resampling**: Reusing the **existing dataset** to draw new samples (with or without replacement) to validate or improve the model.

Think of it like this:

👉 **Sampling** = Getting your first basket of fruits from a farm.

☐ **Resampling** = Taking different combinations of fruits from the basket again and again to check if your fruit salad tastes good every time.



Cross Validation (CV) and resampling

python

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```
import pandas as pd
import numpy as np

# Step 1: Initial Sampling (simulate collecting data)
# Let's say this is our original sample from a population
data = pd.DataFrame({
    'CustomerID': range(1, 11), # 10 customers
    'LoanAmount': [1000, 1500, 800, 1200, 1600, 900, 1400, 1300, 1100, 1700]
})

print("Original Sample (Initial Sampling):")
print(data)

# Step 2: Resampling (Bootstrapping - with replacement)
bootstrap_sample = data.sample(n=10, replace=True, random_state=42)

print("\nResampled Data (Bootstrapping):")
print(bootstrap_sample)
```

Cross Validation (CV) and resampling

Types of Resampling:

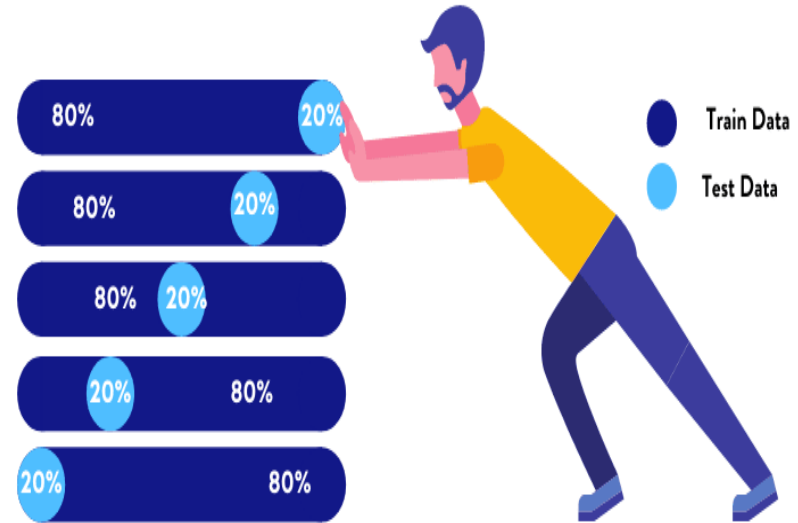
Two common method of Resampling are:

- K-fold **Cross-validation**
- Bootstrapping

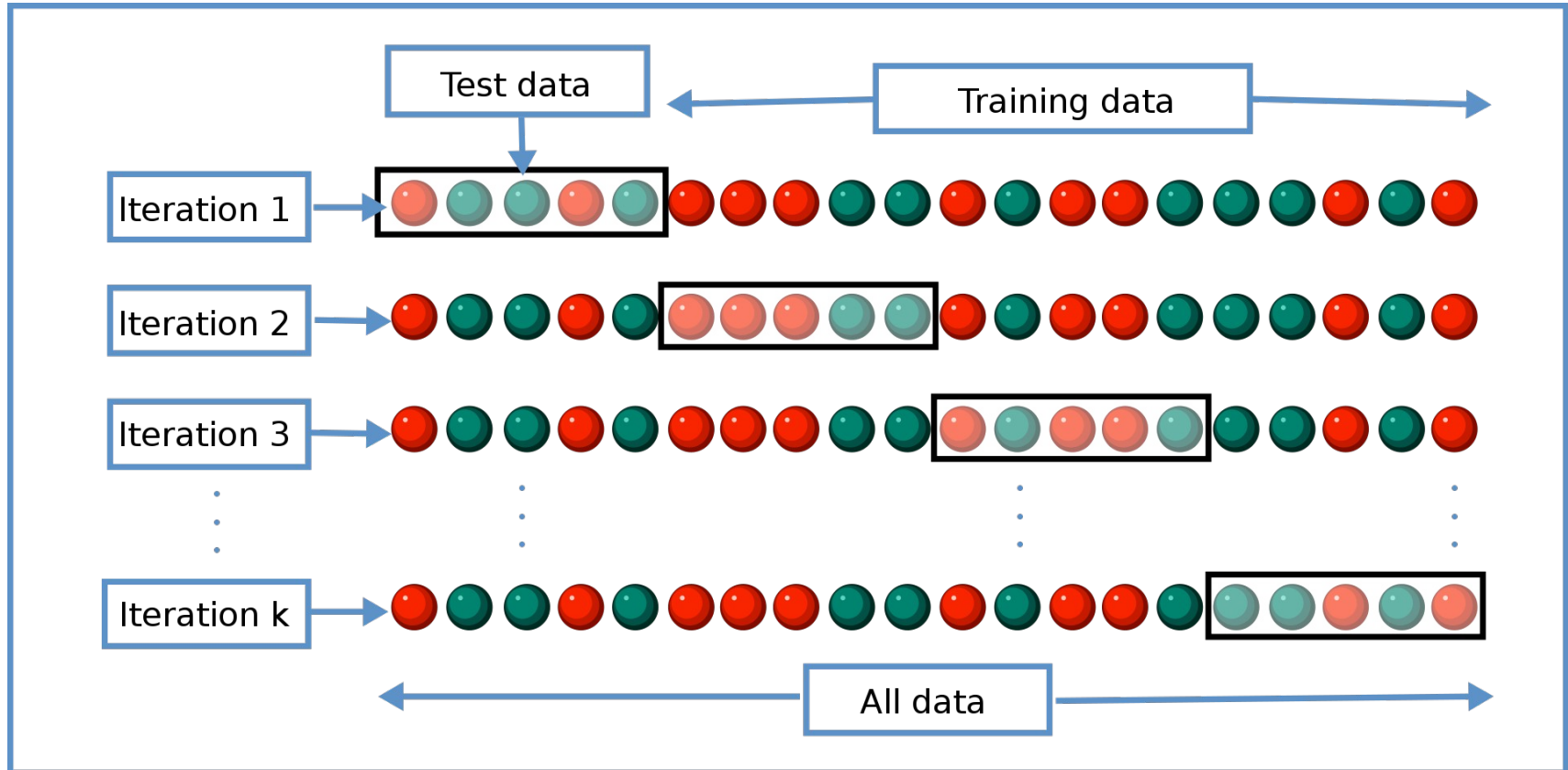
Cross-Validation (CV) is a general technique to evaluate model performance by testing it on different data subsets.

K-Fold CV is a specific type of CV where the dataset is split into k parts; each part gets a turn as the test set.

Real-Time Example: In email spam detection, K-Fold CV ensures the model is tested across different groups of emails to avoid bias from a single test split.



Cross Validation (CV) and resampling



Cross Validation (CV) and resampling

Types of Cross-Validation

1. Holdout: Split data once (e.g., 80% train, 20% test).

☞ *Train on 800 shampoo buyers, test on 200.*

2. K-Fold CV: Split data into K parts, test on each once.

☞ *In 5-fold, test on 200 customers each time, train on 800.*

3. Stratified K-Fold: Like K-Fold but keeps class balance.

☞ *Each fold has equal shampoo+conditioner buyers.*

4. Leave-One-Out (LOOCV): Test on 1 customer at a time.

☞ *Train on 999, test on 1 — repeat 1000 times.*

5. Repeated K-Fold: Run K-Fold CV multiple times randomly.

☞ *Repeat 5-fold CV 3 times to get stable accuracy.*

Cross Validation (CV) and resampling

Types of Cross-Validation

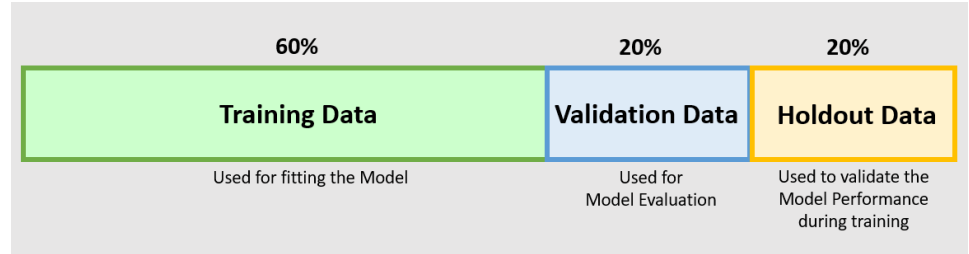
Holdout Validation

- Split the data once into:
 - **Training Set** (e.g., 80%)
 - **Testing Set** (e.g., 20%)
- Train the model on training data and test on testing data.

📦 **Example:** You have sales data from 1000 customers. You use 800 to train a shampoo recommender, and 200 to test it.

👍 Simple and fast

👎 May give biased results if data split is unlucky



Cross Validation (CV) and resampling

Types of Cross-Validation

K-Fold Cross-Validation

Divide the data into **K equal parts (folds)**.

Use one fold for testing, rest for training.

Repeat K times, changing the test fold each time.

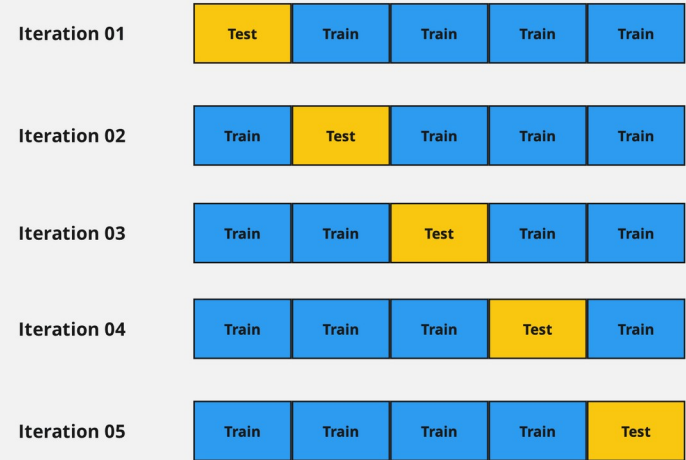
Average the results.

 **Example:** 5-Fold CV on 1000 shampoo buyers
→ split into 5 sets of 200. Train on 800, test on 200 each time.

 More reliable than holdout

 More computation

K-Fold Cross Validation



Cross Validation (CV) and resampling

Types of Cross-Validation

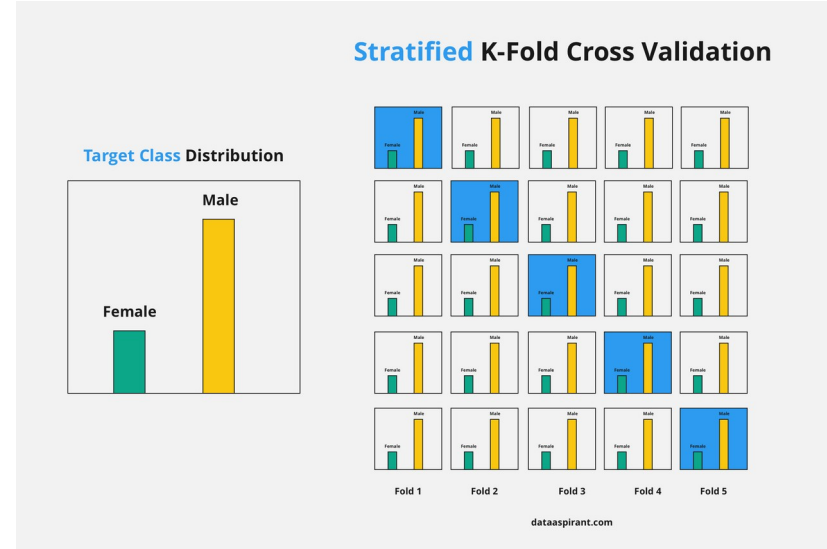
Stratified K-Fold Cross-Validation

Like K-Fold, **but maintains the class balance** (e.g., equal "Yes"/"No" labels) in each fold.

 **Example:** 40% of buyers purchase conditioner. Each fold in Stratified CV will keep 40% conditioner-buyers.

 Best for imbalanced data

 Slightly slower than normal K-Fold



Cross Validation (CV) and resampling

Types of Cross-Validation

Leave-One-Out Cross-Validation (LOOCV)

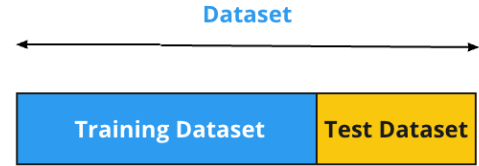
Each data point is used once as a test set.

For **N** data points, run model **N times**.

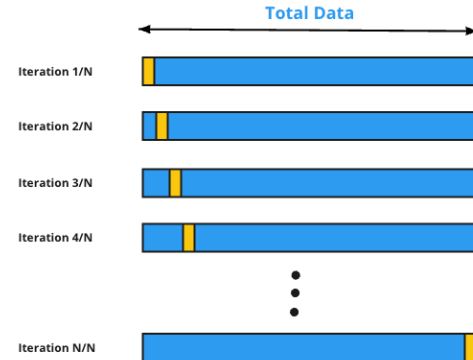
 **Example:** For 10 customer records, train on 9, test on 1 → do this 10 times.

 Very thorough

 Very slow for large datasets



LOOCV: Leave One Out Cross Validation



Cross Validation (CV) and resampling

Types of Cross-Validation

Leave-P-Out Cross-Validation

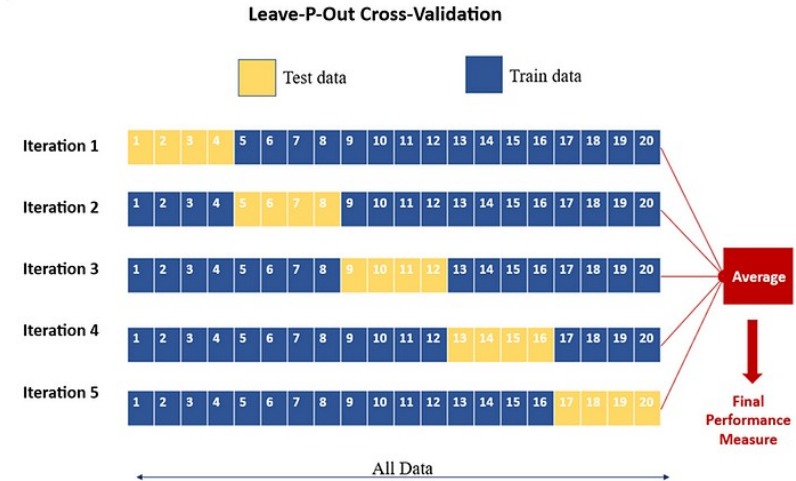
Leave **P** samples out for testing, train on the rest.

Try **all combinations** of P samples.

📦 **Example:** Leave 2 shampoo-buying customers out, train on the rest. Repeat for all possible pairs.

👍 Very detailed

👎 Extremely time-consuming



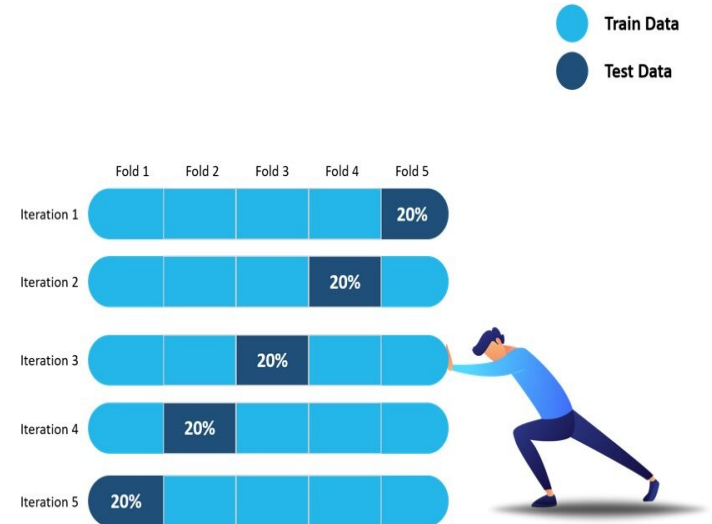
Cross Validation (CV) and resampling

K-fold cross-validation:

K-Fold Cross-Validation splits the data into k equal parts, using one part for testing and the rest for training, repeating the process k times.

This helps ensure the model is tested on every part of the data.

Real-Time Example: In predicting house prices, K-Fold CV allows the model to learn from different neighborhood data and test on unseen areas each time



Cross Validation (CV) and resampling

✓ K-Fold Validation Code

python

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```
from sklearn.model_selection import KFold, cross_val_score
from sklearn.ensemble import RandomForestClassifier
from sklearn.datasets import load_iris
```

```
# Load sample data
```

```
X, y = load_iris(return_X_y=True)
```

```
# Define the model
```

```
model = RandomForestClassifier()
```

```
# Apply K-Fold Cross-Validation (k=5)
```

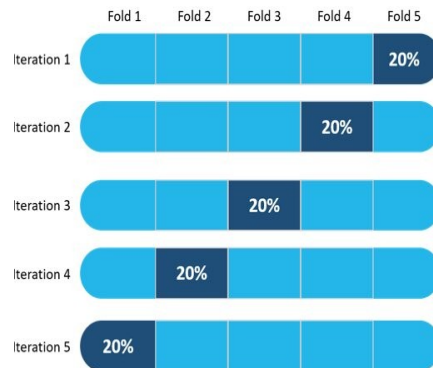
```
kfold = KFold(n_splits=5, shuffle=True, random_state=42)
```

```
scores = cross_val_score(model, X, y, cv=kfold)
```

```
print("K-Fold CV Accuracy Scores:", scores)
```

```
print("Average Accuracy:", scores.mean())
```

● Train Data
● Test Data

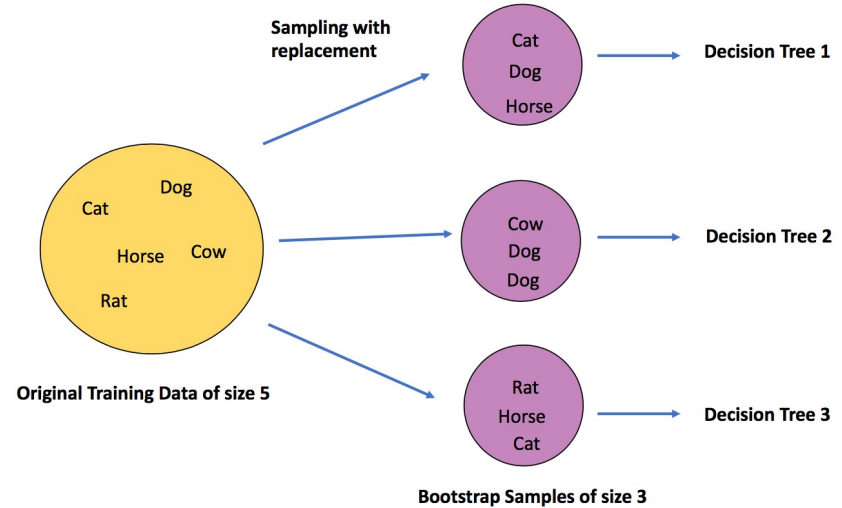


test on unseen areas each time

Cross Validation (CV) and resampling

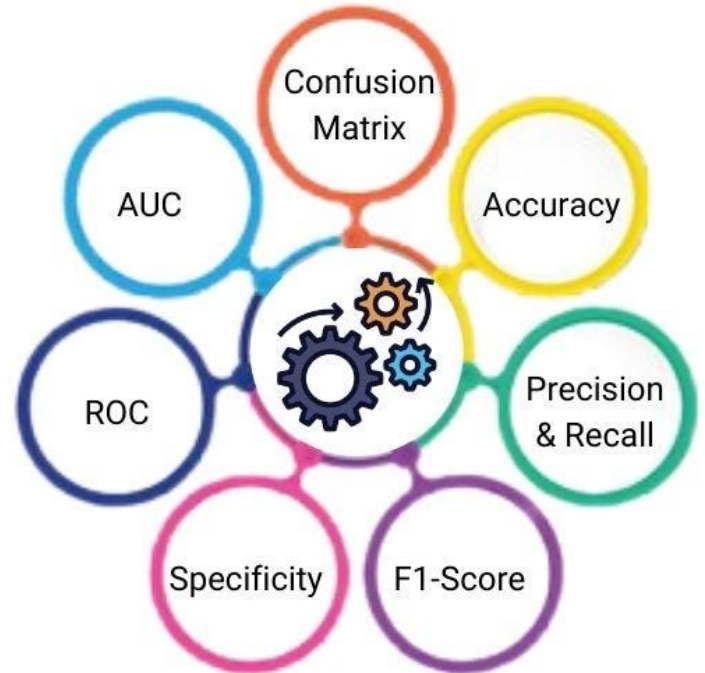
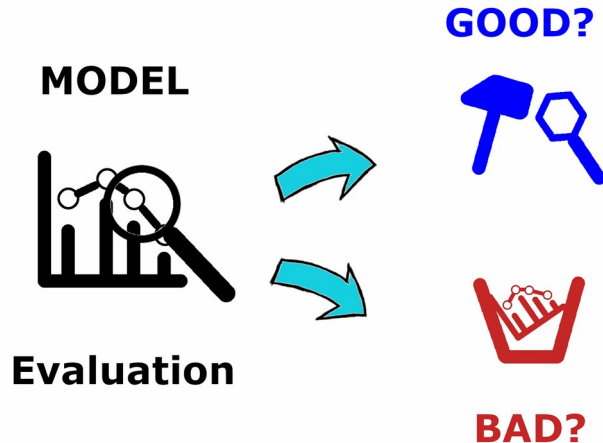
Bootstrapping is a resampling method where data is sampled **with replacement**, meaning the same record can appear multiple times in the training set.

The **left-out data** (not selected in the sample) is used as the **test set**, also called **out-of-bag (OOB)** data.



Measuring Classifier Performance

Measuring Classifier Performance involves evaluating how well a model predicts the correct class labels using metrics like accuracy, precision, recall, and F1 score.

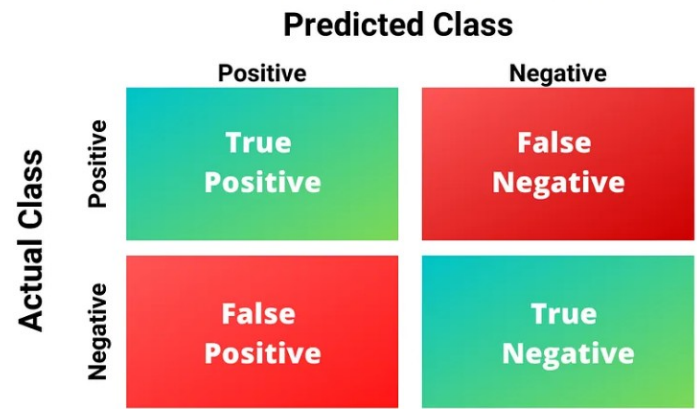


Accuracy and its limitations, Confusion Matrix, Precision & Recall, F1-Score, Specificity, Receiver Operating Characteristic Curve (ROC), Area Under Curve (AUC)

Measuring Classifier Performance

Confusion Matrix

A **Confusion Matrix** shows how many correct and incorrect predictions a classification model made, comparing actual vs predicted labels.



Sample	Actual Value (True Label)	Predicted Value (Model Output)
1	Spam	Spam
2	Spam	Not Spam
3	Not Spam	Not Spam
4	Not Spam	Not Spam
5	Spam	Spam
6	Spam	Spam
7	Not Spam	Spam
8	Not Spam	Not Spam
9	Not Spam	Not Spam
10	Spam	Not Spam

Measuring Classifier Performance

		Predicted Class	
		Positive	Negative
Actual Class	Positive	True Positive	False Negative
	Negative	False Positive	True Negative

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

Sample	Actual Value (True Label)	Predicted Value (Model Output)
1	Spam	Spam
2	Spam	Not Spam
3	Not Spam	Not Spam
4	Not Spam	Not Spam
5	Spam	Spam
6	Spam	Spam
7	Not Spam	Spam
8	Not Spam	Not Spam
9	Not Spam	Not Spam
10	Spam	Not Spam

Measuring Classifier Performance

Accuracy

- **Definition:** Measures overall correctness of the model.
- **Formula:**

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Calculation:**

$$\frac{3 + 4}{3 + 4 + 1 + 2} = \frac{7}{10} = 0.70 \text{ or } 70\%$$

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	TP = 3	FN = 2
Actual: Not Spam	FP = 1	TN = 4

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

Measuring Classifier Performance

Precision

- **Definition:** Out of all predicted spams, how many were actually spam.
- **Formula:**

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **Calculation:**

$$\frac{3}{3 + 1} = \frac{3}{4} = 0.75 \text{ or } 75\%$$

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	TP = 3	FN = 2
Actual: Not Spam	FP = 1	TN = 4

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

Measuring Classifier Performance

◆ Precision for Spam (Positive Class)

This is what we usually calculate:

$$\text{Precision}_{\text{Spam}} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

It tells: "Of all emails predicted as Spam, how many were actually Spam?"

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	TP = 3	FN = 2
Actual: Not Spam	FP = 1	TN = 4

◆ Precision for Not Spam (Negative Class)

You can also calculate:

$$\text{Precision}_{\text{Not Spam}} = \frac{\text{True Negatives}}{\text{True Negatives} + \text{False Negatives}}$$

It tells: "Of all emails predicted as Not Spam, how many were actually Not Spam?"

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

Measuring Classifier Performance

Recall (Sensitivity)

- **Definition:** Out of all actual spam emails, how many did the model correctly predict.
- **Formula:**

$$\text{Recall} = \frac{TP}{TP + FN}$$

- **Calculation:**

$$\frac{3}{3 + 2} = \frac{3}{5} = 0.60 \text{ or } 60\%$$

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	TP = 3	FN = 2
Actual: Not Spam	FP = 1	TN = 4

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

Measuring Classifier Performance

Recall (Sensitivity)

🔴 Recall Formulas:

- Recall (Spam) =

$$\frac{\text{True Positives (TP)}}{\text{True Positives} + \text{False Negatives}} = \frac{3}{3 + 2} = 0.60$$

- Recall (Not Spam) =

$$\frac{\text{True Negatives (TN)}}{\text{True Negatives} + \text{False Positives}} = \frac{4}{4 + 1} = 0.80$$

- Definition: Average of recall values for each class.

- Formula:

$$\text{Macro Recall} = \frac{\text{Recall}_{\text{spam}} + \text{Recall}_{\text{not spam}}}{2}$$

- From earlier:

$$\frac{0.60 + 0.80}{2} = 0.70$$

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	TP = 3	FN = 2
Actual: Not Spam	FP = 1	TN = 4

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

Measuring Classifier Performance

F1 Score

- **Definition:** Harmonic mean of precision and recall.
- **Formula:**

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Calculation:**

$$\frac{2 \times 0.75 \times 0.60}{0.75 + 0.60} = \frac{0.90}{1.35} \approx 0.667 \text{ or } 66.7\%$$

	Predicted: Spam	Predicted: Not Spam
Actual: Spam	TP = 3	FN = 2
Actual: Not Spam	FP = 1	TN = 4

- ❑ **False Negatives (FN)** = 2 → Samples 2, 10
- ❑ **True Negatives (TN)** = 4 → Samples 3, 4, 8, 9
- ❑ **True Positives (TP)** = 3 → Samples 1, 5, 6
- ❑ **False Positives (FP)** = 1 → Sample 7

t test, McNemar's test, K-fold CV paired t test

The **T-test**, **McNemar's test**, and **K-fold CV paired t-test** are used to evaluate and clarify the results of hyperparameter tuning, ensuring that any observed differences in model performance are statistically significant.

		Model 2	
		correct	wrong
Model 1	correct	9959	11
	wrong	1	29

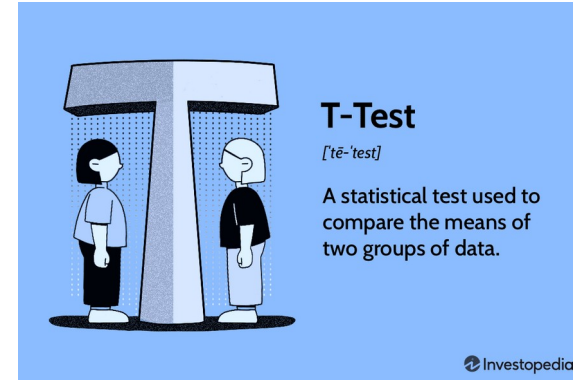
		Model 2	
		correct	wrong
Model 1	correct	9945	25
	wrong	15	15

t-Test

1. t-Test:

- **Purpose:** Compares the **means** of **two independent groups** to see if they are significantly different.
- **Example:** Comparing **test scores** of two different classes (Class A vs Class B).

A **t-test** (specifically the **independent t-test**) can indeed be used to **compare the performance of the same algorithm on two different datasets**.



t-Test

1. t-Test:

A **t-test** (specifically the **independent t-test**) can indeed be used to **compare the performance of the same algorithm on two different datasets**.

- Half of give significance level is considered
- T-test table used for table value

FORMULA:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

$$\text{where } S = \sqrt{\frac{\sum x_1^2 + \sum x_2^2}{n_1 + n_2 - 2}}$$

$$\gamma^2 = n_1 + n_2 - 2$$

t-Test

1. t-Test:

The nicotine contents in two random samples of tobacco are given below.

Sample I 21 24 25 26 27

Sample II 22 27 28 30 31 36

Can you say that the two samples came from the same population ?

X_1	$x_1 = (X_1 - \bar{X}_1)$	x_1^2	X_2	$x_2 = (X_2 - \bar{X}_2)$	x_2^2
21	-4	16	22	-7	49
24	-1	1	27	-2	4
25	0	0	28	-1	1
26	1	1	30	1	1
27	2	4	31	2	4
			36	7	49
Σ		22	174		108
$\Sigma X_1 = 123$			$\Sigma X_2 = 174$		

FORMULA:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

where $S = \sqrt{\frac{\Sigma x_1^2 + \Sigma x_2^2}{n_1 + n_2 - 2}}$

$$y = n_1 + n_2 - 2$$

t-Test

1. t-Test:

$$\therefore S = \sqrt{\frac{\sum x_1^2 + \sum x_2^2}{n_1 + n_2 - 2}}$$

$$S = \sqrt{\frac{22 + 108}{5 + 6 - 2}}$$

$$S = 3.8$$

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

$$t = \frac{25 - 29}{3.8} \sqrt{\frac{5 \times 6}{5 + 6}}$$

$$t = \frac{-4}{3.8} \times 1.651$$

FORMULA:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{S} \sqrt{\frac{n_1 n_2}{n_1 + n_2}}$$

$$\text{where } S = \sqrt{\frac{\sum x_1^2 + \sum x_2^2}{n_1 + n_2 - 2}}$$

$$\nu = n_1 + n_2 - 2$$

$$\therefore t = \underline{1.738} \text{ (ignore sign)}$$

$$\nu = n_1 + n_2 - 2$$

$$\nu = 5 + 6 - 2$$

$$\nu = 9$$

X_1	$x_1 = (X_1 - \bar{X}_1)$	x_1^2	X_2	$x_2 = (X_2 - \bar{X}_2)$	x_2^2
21	-4	16	22	-7	49
24	-1	1	27	-2	4
25	0	0	28	-1	1
26	1	1	30	1	1
27	2	4	31	2	4
			36	7	49
Σ		22	174		108
$\Sigma X_1 = 123$			$\Sigma X_2 = 174$		

t-Test

1. t-Test:

$$\therefore t = \underline{1.738} \text{ (ignore sign)}$$

$$\nu = n_1 + n_2 - 2$$

$$\nu = 5 + 6 - 2$$

$$\nu = 9$$

Since no significance level is given, 5% taken as default.

For t-test, $\frac{5\%}{2} = 2.5\%$ level of significance considered.

$$\therefore t_{0.025} = 2.262$$

Critical values of t for one-tailed tests

Significance level (α)

Degrees of freedom (df)	.2	.15	.1	.05	.025	.01	.005	.001
1	1.376	1.963	3.078	6.314	12.706	31.821	63.657	318.309
2	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327
3	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215
4	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173
5	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144
11	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025
12	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930
13	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852
14	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787
15	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733
16	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686
17	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646
18	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610
19	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579
20	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552

t-Test

1. t-Test:

! Why Not Significant?

To say there is a significant difference, your t-value must be more than the critical value:

Check: $|t| = 2.03 < 2.262 =$ Not enough evidence

So, even though the means are different, the difference is not strong enough when we consider the variability (spread) in the data.

T-test

🎓 Example:

Suppose you test two teaching methods:

- Group A score average = 80
- Group B score average = 85
- You do a t-test $\rightarrow t = 3.0$
- Critical $t = 2.262 \rightarrow 3.0 > 2.262 \rightarrow \checkmark$ Significant!

So you conclude:

"Teaching Method B works better than A."

📌 Summary:

Situation	What It Means	What You Do
$t < \text{critical value}$	Not significant	Accept null: groups are similar
$t \geq \text{critical value}$	Significant	Reject null: groups are different

Critical values of *t* for one-tailed tests

Significance level (α)

Degrees of freedom (<i>df</i>)	.2	.15	.1	.05	.025	.01	.005	.001
1	1.376	1.963	3.078	6.314	12.706	31.821	63.657	318.309
2	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327
3	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215
4	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173
5	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893
6	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208
7	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785
8	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501
9	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297
10	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144
11	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025
12	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930
13	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852
14	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787
15	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733
16	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686
17	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646
18	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610
19	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579
20	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552

Critical values of *t* for one-tailed tests

Significance level (α)

Degrees of freedom (<i>df</i>)	.2	.15	.1	.05	.025	.01	.005	.001
21	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527
22	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505
23	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485
24	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467
25	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450
26	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435
27	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421
28	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408
29	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396
30	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385
40	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307
50	0.849	1.047	1.299	1.676	2.009	2.403	2.678	3.261
60	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232
70	0.847	1.044	1.294	1.667	1.994	2.381	2.648	3.211
80	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195
100	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174
1000	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098
Infinite	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090

McNemar's Test:

2. McNemar's Test:

- **Purpose:** Compares **binary outcomes** (yes/no) between **two related groups** to see if there's a significant change.
- **Example:** Comparing **pre-treatment vs post-treatment** success rates in the same patients.
- **Purpose:** It's used to compare the performance of **two different classification algorithms on the same dataset** by analyzing **binary outcomes** (like "Yes" or "No", "True" or "False").
- It helps determine if there is a **statistically significant difference** in the performance of two models when applied to the same dataset.

McNemar's Test:

Purpose:

McNemar's test is a **statistical test** used on **paired nominal data**. It checks if **two classifiers (or treatments)** disagree in a significant way when tested on the **same dataset**.

■ When to Use It?

Use McNemar's test when:

- You're comparing **two classifiers** on the **same samples**.
- The outcome is **binary** (e.g., correct vs incorrect, yes vs no).
- You want to know if the **difference in their performance** is statistically significant.

	model 2 correct	model 2 wrong
model 1 correct	A	B
model 1 wrong	C	D

McNemar's Test:

Contingency Table (2x2)

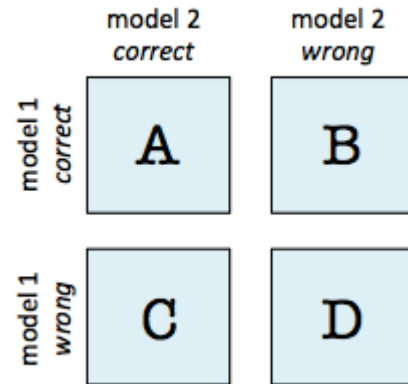
	Classifier B Correct	Classifier B Incorrect
Classifier A Correct	a (both correct)	b (A correct, B wrong)
Classifier A Incorrect	c (A wrong, B correct)	d (both wrong)

- **b** = Number of samples where **A** was **correct**, but **B** was **wrong**
- **c** = Number of samples where **A** was **wrong**, but **B** was **correct**

McNemar's Test Formula (without continuity correction):

$$\chi^2 = \frac{(b - c)^2}{b + c}$$

If **b + c** ≥ 25 , you can use this formula directly.



McNemar's Test:

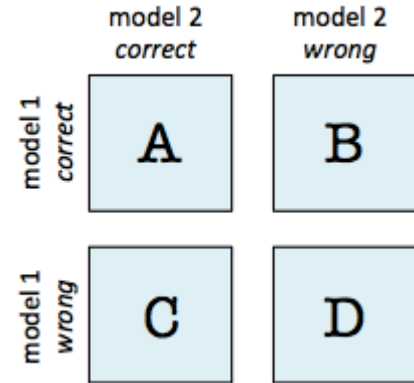
Example Scenario:

Suppose 100 people try both Shampoo and Conditioner and report whether each worked for them (Yes = Effective, No = Not Effective).

	Conditioner Effective	Conditioner Not Effective
Shampoo Effective	40	20
Shampoo Not Effective	10	30

This gives us the following contingency table:

	Conditioner Yes	Conditioner No	Total
Shampoo Yes	40	20	60
Shampoo No	10	30	40
Total	50	50	100



McNemar's Test:

✓ Focus of McNemar's Test:

We focus only on disagreeing pairs:

- **b** = Shampoo Yes, Conditioner No = 20
- **c** = Shampoo No, Conditioner Yes = 10

📊 McNemar's Formula:

$$\chi^2 = \frac{(b - c)^2}{b + c}$$
$$\chi^2 = \frac{(20 - 10)^2}{20 + 10} = \frac{100}{30} = 3.33$$

🔧 Decision Rule:

- Degrees of freedom = 1
- At significance level $\alpha = 0.05$, critical value from chi-square table = 3.84

Since $3.33 < 3.84$, we fail to reject the null hypothesis.

	model 2 correct	model 2 wrong
model 1 correct	A	B
model 1 wrong	C	D

Chi-square Test – Distribution table and formulas

Degrees of Freedom (df) Significance Level (α)	0.01	0.05	0.10	0.25	0.50
1	6.635	3.841	2.706	1.323	0.454
2	9.210	5.991	4.605	2.773	1.386
3	11.345	7.815	6.251	3.930	2.366
4	13.277	9.488	7.779	5.178	3.357
5	15.086	11.070	9.236	6.571	4.351
6	16.812	12.592	10.645	7.962	5.348
7	18.475	14.067	12.017	9.364	6.346
8	20.090	15.507	13.362	10.773	7.344
9	21.666	16.919	14.684	12.189	8.343
10	23.209	18.307	15.987	13.603	9.342

Paired T-test

the **Paired T-test** is used in **both classification and regression problems**, but:

- In **classification**, it compares **metrics** like **accuracy, precision, recall, or F1-score** from two models over multiple data splits (e.g., K-fold cross-validation).
- In **regression**, it compares **metrics** like **RMSE, MAE, or R^2** from two models over the same data splits.
- the **Paired T-test** does **not use different datasets** — it **requires the same dataset** split in the same way for both models.

Paired T-test

Compare **Model A** and **Model B** (same algorithm, different hyperparameters) using **5-Fold Cross-Validation**, and check if one is significantly better using the **paired t-test**.

 **Setup:**

Fold	Model A Accuracy	Model B Accuracy	Difference (A - B)
1	0.85	0.88	-0.03
2	0.82	0.86	-0.04
3	0.80	0.84	-0.04
4	0.83	0.87	-0.04
5	0.81	0.85	-0.04

Paired T-test

Step-by-Step Paired T-test Calculation:

1. Mean difference ($\bar{d}$):

$$\text{Mean} = \frac{-0.03 + (-0.04) + (-0.04) + (-0.04) + (-0.04)}{5} = -0.038$$

2. Standard deviation (s) of the differences:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (d_i - \bar{d})^2}$$

• Differences from mean:

- $(-0.03 + 0.038)^2 = 0.000064$
- $(-0.04 + 0.038)^2 = 0.000004$ (3 times)

• Sum of squares: $0.000064 + 3 \times 0.000004 = 0.000076$

$$s = \sqrt{\frac{0.000076}{4}} = \sqrt{0.000019} \approx 0.00436$$

3. T-statistic:

$$t = \frac{\bar{d}}{s/\sqrt{n}} = \frac{-0.038}{0.00436/\sqrt{5}} \approx \frac{-0.038}{0.00195} \approx -19.49$$

Paired T-test

 Critical t-value ($df = 4$, $\alpha = 0.05$):

- From t-table: t-critical ≈ 2.776

 **Conclusion:**

Since $|t| = 19.49 > 2.776$, the difference is **statistically significant**.

Model B is significantly better than Model A based on cross-validated accuracy.